Daily Drilling Report
SPD19B-03
8/30/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
34.0	12 1/4"	179.00	72.83	31.70	94.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
13 3/8" Shoe, 1,487.00mKB	225,147.73	8,941.57	7,948,216.10		29.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,496.00	2,600.00		104.00	14.90	7.0

Operations at Report Time
Drilling @ 2631m.

Operations Summary
Drilling 12 1/4" hole section f/ 2496m to 2600m.

Operations next 24hrs
Cont' drilling 12 1/4" hole section.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Behnam Jahangiri / Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	8/16/2014	9/16/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	6/16/2014	9/15/2014
Pit Drill	8/25/2014	10/10/2014
Trip Drill	8/28/2014	10/18/2014

STOP Cards

Number of Reports: 35

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	03:00	3.00	12 1/4"	D	07	N	Increase MW f/ 11.4 ppg to 11.6 ppg and treat the mud. Meanwhile keep the string in motion w/ 30 rpm / 700 gpm. Bring the mud properties to the normal range.
03:00	00:00	21....	12 1/4"	D	03	P	Resume directional drilling 12 1/4" hole in rotation mode f/ 2496m to 2600m w/ 800 gpm/ 2100-2200 psi/ 37-40 kips/ 60-40 rpm/ 14-23 kft-lbf. ** Drilling with NL/NG. ** Pump 20 bbl Hi-Vis pill when STD down. ** B/R and ream down each stand drilled once. ** Take survey before connection. >> Drilling Practice in "S8" Member (2498.7m - 2526.8m): ** For the first 10m in "S8", back ream gently to top of S8 (2498.7m) every 1m drilled. ** For the rest of drilling in S8, B/R gently to top of S8 (2498.7m) every 5m drilled till the base of S8. ** At stand down in "S8", pump 30 bbls Lo-vis + 30 bbls Hi-Vis Pill. ** B/R & R/D precautionary the entire stand twice w/ 700 gpm, 30 rpm. ** Spot 20 bbls of ±15 ppg pill on bottom prior to connection at the end of stand. ** Skip survey in S8.
00:00	00:00						Aug 31, 2014 00:00 - 06:00 Con't directional drilling 12 1/4" hole section f/ 2600m to 2631m w/ 800 gpm/ 2100-2250 psi/ 40 kips/ 40 rpm/ 15-23 kft-lbs. ** Last Sample @ 2620m: Anhy (20%) + Chalky Lst (40%) + Dol. (30%) + Clst (10%)

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Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,600.00	KCl-Polymer Mud	11.70	62	25.0	31.000	3.5
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.0	0.100	1.400	0.900			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
7	10	2.0	76.0	22.0	55,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
11.000	20.000	25.000	2.6/97.4	70.0	2066.0	1560.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
5107.0	149.0	0.0	0.01		17.8866	0.425

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
1.00	6.	180	2.0	999.0	33.7

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Mataf Pride	Crew Boat	8/30/2014 21:05	8/30/2014 21:10	Disembarked 01 PAX and Embarked 01 PAX for Kish, departed for Kish.
Swissco	Supply Boat	8/31/2014 02:30		A/S DD6 Delivering cargo from Aban 8
Forte	Supply Boat	8/29/2014 19:30	8/30/2014 10:00	Del DD6 Cargo & Cement, B/L Cargo for Kish / SHJ, Dep for Kish via Sahar 1
Marina-1	Supply Boat	8/16/2014 15:15		Standby at Anchorage.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Caustic Soda 2014	15.0	340.0	192.0
Citric Acid BG	10.0	150.0	124.0
Starch	160.0	1,080.0	274.0
Pac-LV	20.0	422.0	155.0
PTS (Temp. Stabilizer)	1.0	19.0	17.0
Soda Ash 25 Kg	18.0	550.0	291.0
Sodium Bicarbonate	10.0	230.0	205.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	190.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0
Cement G	0.0	47.0	179.0
Diesel Fuel	9.0	0.0	163.0
Drill Water	9.0	0.0	314.0
Potable Water	20.0	20.0	102.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
9	12 1/4	Varel	VM616PGUX-V2	6011971	PDC	___	12/12/12/12/12/12/12/12/12
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.99	28.00	250,281.60	2,437.00	2,600.00	163.00	5.8	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
750	810	1,650.0	2,210.0	7	40	125	157
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
39.84	28.00	100.00	0.00	388	350	296	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,496.00	2,524.40	4.90	21,000.0	805	60	2,200.0	38	343
2,524.40	2,553.20	4.00	19,000.0	800	60	2,150.0	40	346
2,553.20	2,582.00	3.70	19,000.0	800	60	2,200.0	37	346
2,582.00	2,600.00	2.30	23,000.0	800	40	2,210.0	38	350

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.000	27.83	231.39
7" Drilling Jar (COR-3022-AD)	7	2.938	5.85	203.56
16 jts 5 7/8" HWDP	5 7/8	4.000	149.23	197.71
X/O	7	2.813	0.79	48.48
1 jt 8 1/4" DC	8 1/4	2.810	9.46	47.69
8 1/2" Float Sub	8 1/2		0.79	38.23
8" NMDC	8	3.500	9.44	37.44
8" TM - Sub	8		2.03	28.00
8" NMDC	8	3.500	9.46	25.97
11 7/8" S. Stabilizer (COR3103AD)	11 7/8	2.813	2.16	16.51
8" PNMDC	8 1/16	3.500	4.03	14.35
9 5/8" Motor (AKO: 1.00° , Sleeve: 12 1/8")	9 5/8		9.95	10.32
12 1/4" PDC bit	12 1/4		0.37	0.37

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
1	2,493.00	14.00	20	185.0	97	114

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No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
1	2,493.00	14.00	30	240.0	97	170
1	2,493.00	14.00	40	315.0	97	227
2	2,493.00	14.00	30	250.0	97	170
2	2,493.00	14.00	40	325.0	97	227
2	2,493.00	14.00	20	195.0	97	114

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
2,529.70	247.60	20.70	2,397.69	-315.33	-682.67	751.97	0.06	0.00
2,557.88	247.90	20.80	2,424.04	-319.11	-691.91	761.95	0.16	0.11
2,587.38	248.00	21.00	2,451.60	-323.06	-701.66	772.46	0.21	0.20

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
S8		2,498.70	2,368.67
S7		2,526.80	2,394.98
S6		2,552.80	2,419.29
S5		2,571.60	2,436.86
S4		2,594.00	

General Remarks

Comment

>> Requirements: (70 BB CaCo3 f/ MI MR #13).

>>> Sent:
** Forte to Kish: (HPOGC 9 5/8" Rotor & lithium metal batteries) + (ABAN empty food containers) + (Almasaoud lifting gear inspection tools) + (Garbage)

>>> Received:
** Forte f/ Kish: (47 MT bulk CMT "G" class)
** Swisco Supplier f/ ABAN 8: (69 jts of 3 1/2" S-135 drill pipes) + (2 x ABAN Cylinder racks) + (6 x ABAN 6 1/2" piston)